

Computer Communication & Networks

Lecture 20

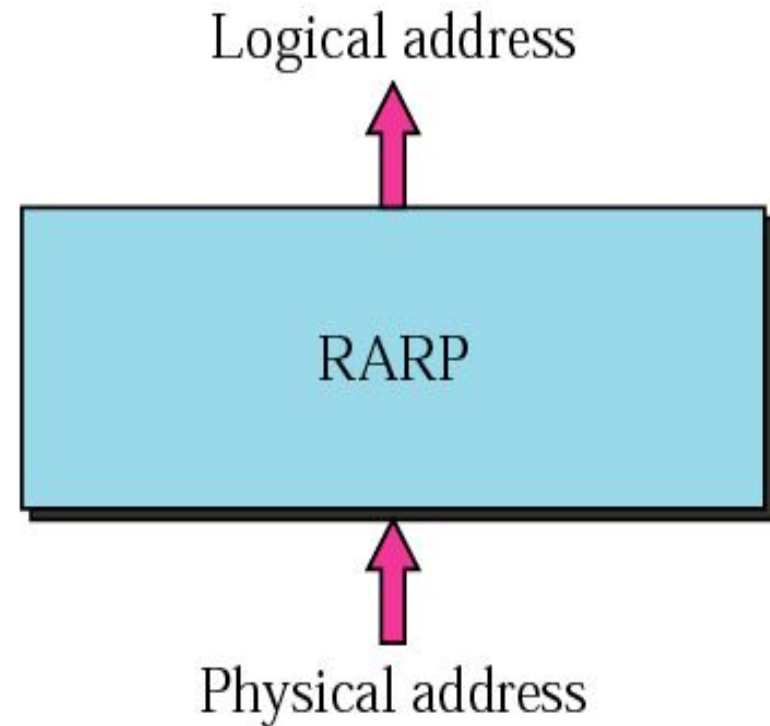
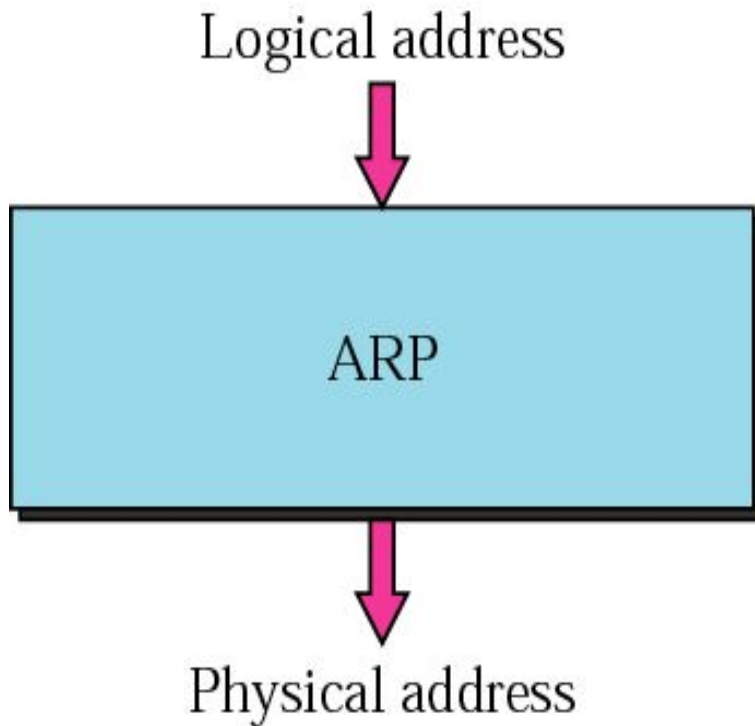
Network Layer: IP and Address Mapping (contd.)

Address Mapping

Address Mapping

- The delivery of a packet to a host or a router requires two levels of addressing: **logical** and **physical**. We need to be able to map a logical address to its corresponding physical address and vice versa. This can be done by using either static or dynamic mapping.

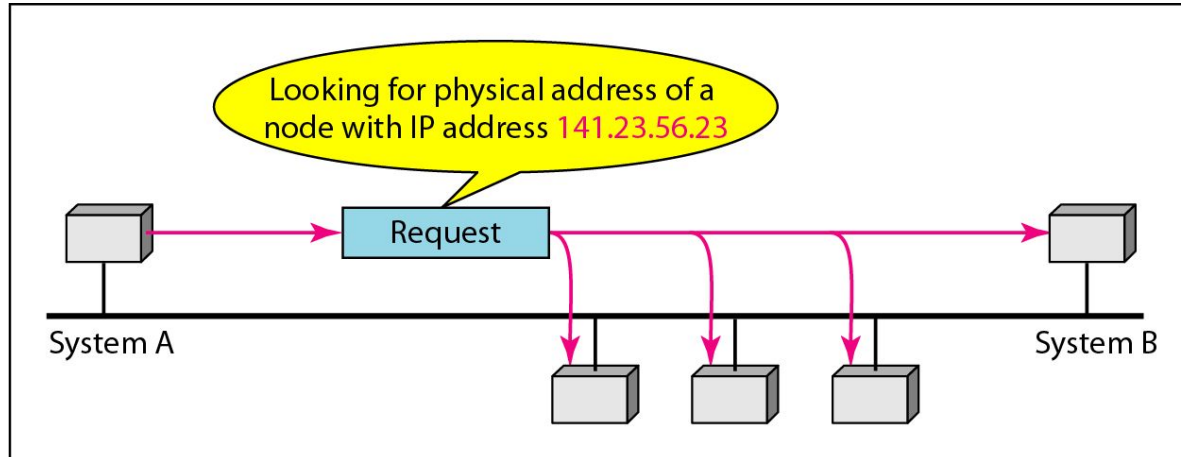
ARP and RARP



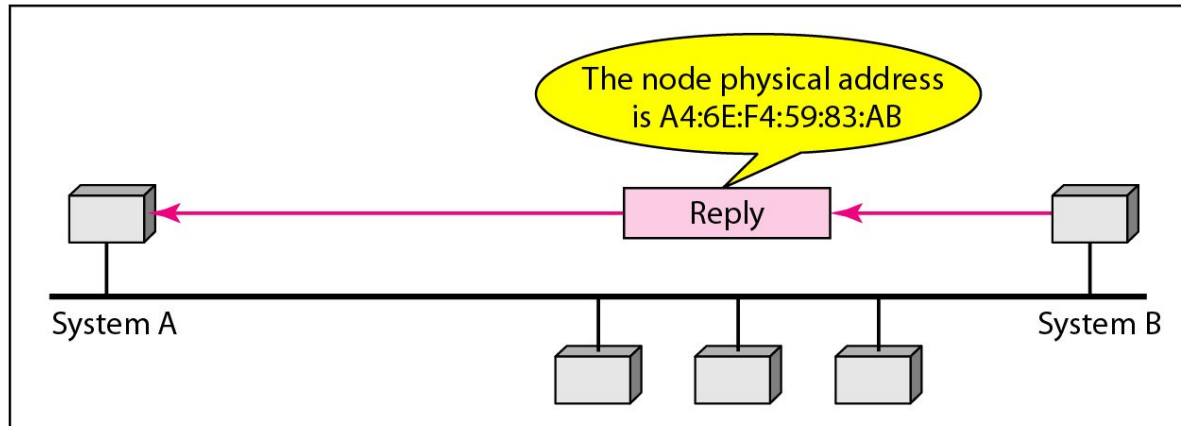
ARP - The Address Resolution Protocol

- An IP datagram must be encapsulated in a frame to pass through the physical network. This requires the physical address of the receiver.
- ARP associates an IP address with its physical address.
- When the physical address of another host is required an ARP query packet is sent which includes
 - IP address of sender
 - IP address of the receiver
- The intended recipient recognises its IP address and send back an ARP response which contains the physical address.

ARP operation

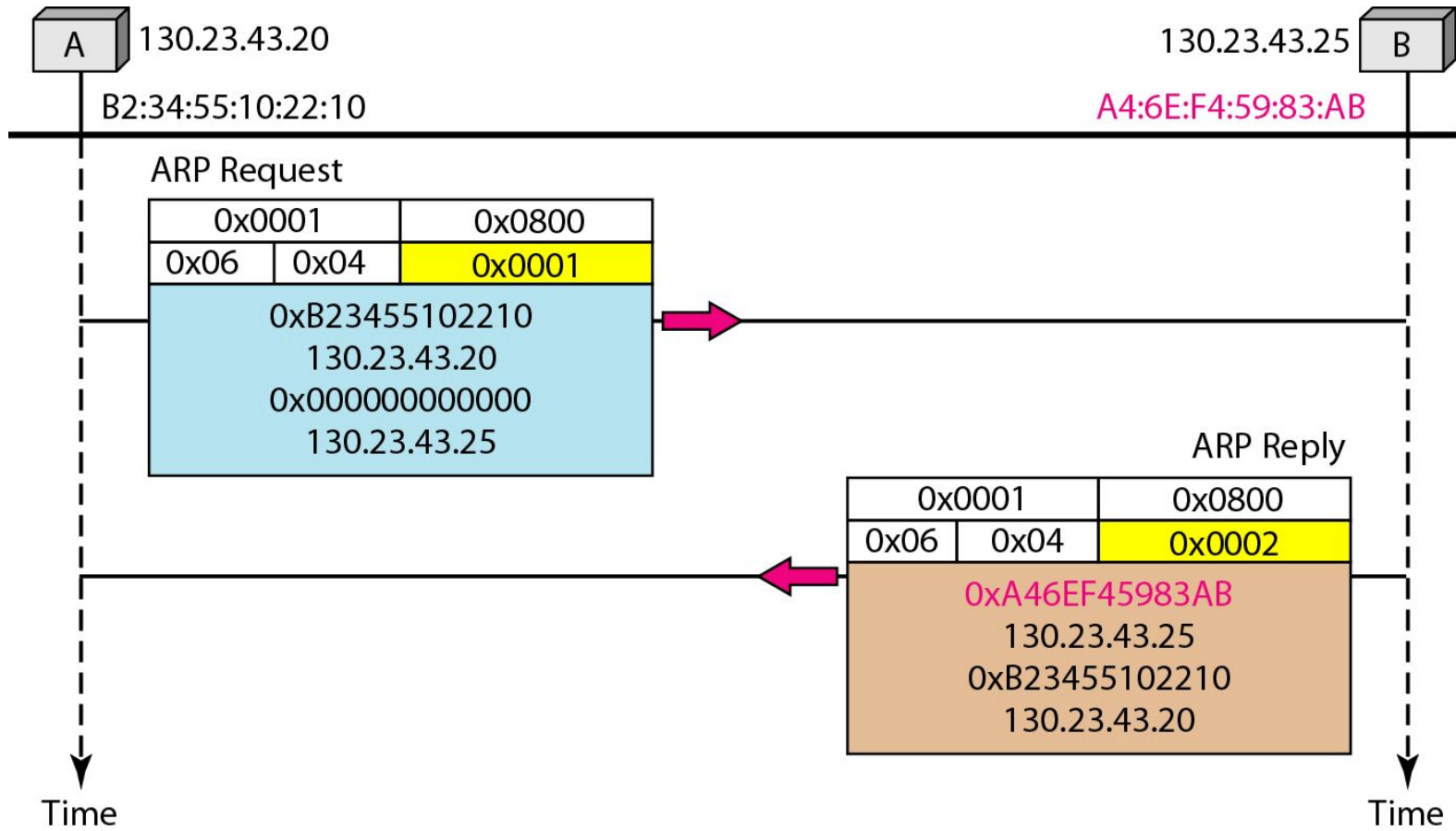


a. ARP request is broadcast

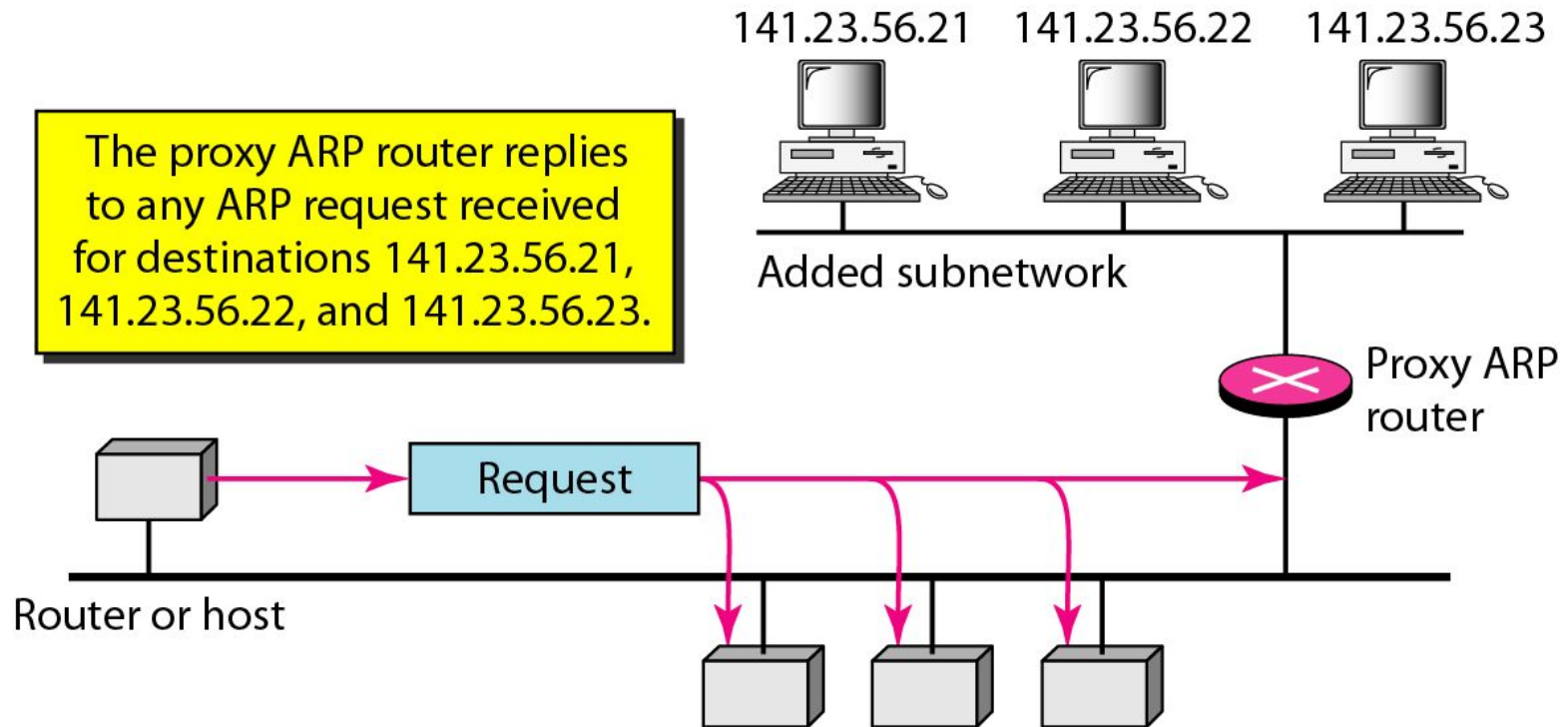


b. ARP reply is unicast

ARP Request & Reply



Proxy ARP



ARP functions

- Sender knows IP address
- IP asks ARP to create an ARP request message
 - sender physical address
 - sender IP address
 - target IP address
 - target physical address filled with 0's
- Message passed to data link layer; destination address = broadcast address
- All hosts receive frame; only target host accepts the packet
- Target host replies (unicast) with an ARP reply message containing the physical address
- Sender receives reply; it now knows the physical address of the target host
- IP datagram's are now unicast to the destination host

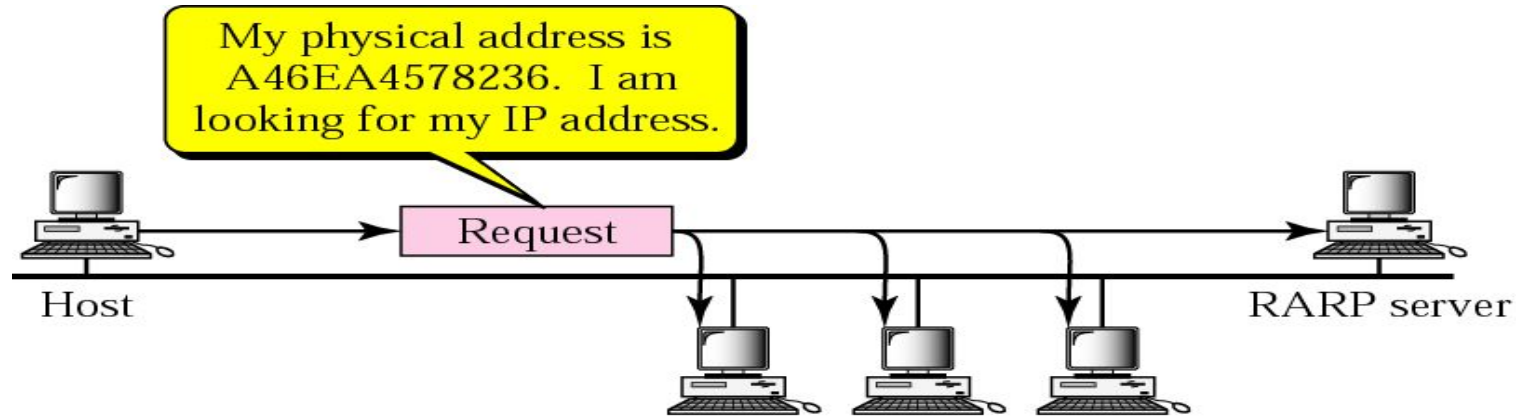
Cache Table

- A sender usually has more than one IP datagram to send to the same destination
- It is inefficient to use ARP for each datagram
- A cache table is used
 - limited size
 - mappings retained only for a limited time

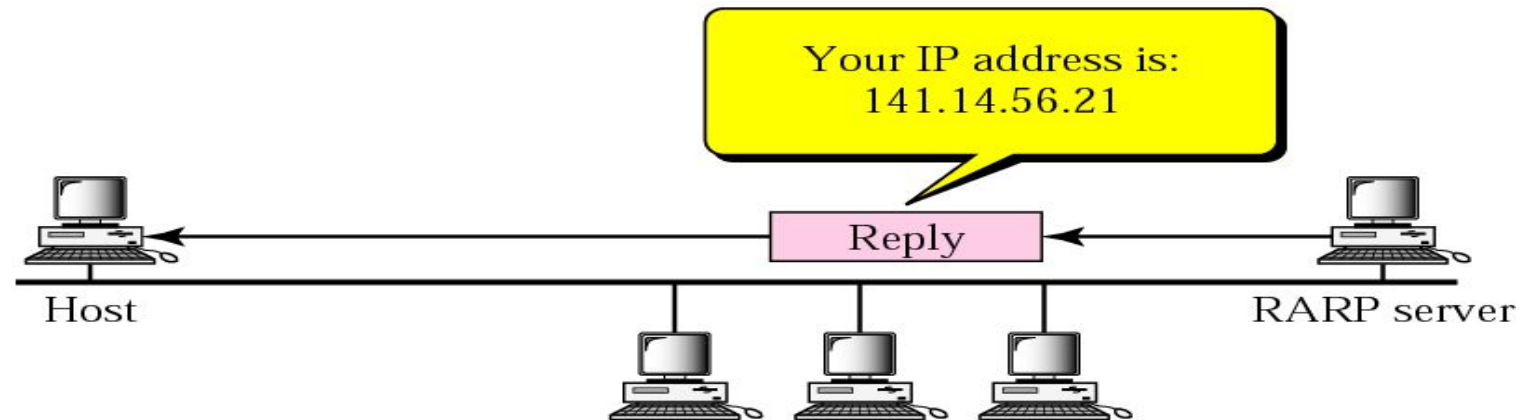
RARP

- Finds the logical address for a host that only knows its physical address
- RARP request packets are broadcast; RARP reply packets are unicast
- Used by diskless machines to obtain their IP addresses

Operation



a. RARP request is broadcast



b. RARP reply is unicast

Note

**An ARP request is broadcast;
an ARP reply is unicast.**

ICMP

ICMP protocol

- IP protocol is a best-effort delivery service, however it has two deficiencies
 - Lack of error control
 - Lack of assistance mechanisms
- IP protocol has no error-reporting or error-correction mechanism
 - What happens when something goes wrong?
 - What happens if a router must discard a datagram because it cannot find a route to the final destination?
 - What if the time-to-live field has a zero value?
 - What if it has to discard all fragments because not all were received in a predetermined time limit?
- IP protocol also lacks a mechanism for host and management queries.
- ICMP was designed to compensate for these deficiencies.

Internet Control Message Protocol (ICMP)

ICMP is used by routers for exchanging error information and for testing. The principal ICMP message types are shown below.

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo request	Ask a machine if it is alive
Echo reply	Yes, I am alive
Timestamp request	Same as Echo request, but with timestamp
Timestamp reply	Same as Echo reply, but with timestamp

Internet Control Message Protocol (ICMP)

- Encapsulated in IP packet
 - Handles error and control messages
 - If router cannot deliver or forward a packet, it sends an ICMP “host unreachable” message to the source
 - If router receives packet that should have been sent to another router, it sends an ICMP “redirect” message to the sender; Sender modifies its routing table
 - ICMP “router discovery” messages allow host to learn about routers in its network and to initialize and update its routing tables
 - ICMP echo request and reply facilitate diagnostic and used in “ping”
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Example – Echo request

The screenshot shows the 'pingtesla - Ethereal' window with a network traffic capture. The main pane displays a list of packets. Packet 4 is selected, showing an ICMP Echo (ping) request from 192.168.2.18 to 128.100.11.1. The details pane below shows the structure of this packet: Ethernet II, Internet Protocol, and Internet Control Message Protocol (Type: 8, Code: 0). The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Info
1	0.000000	00000000.0001031dccb7	00000000.Broadcast	IPX SAP	Nearest Query
2	13.526454	192.168.2.18	192.168.2.1	DNS	Standard query A tesla.comm.utoronto.ca
3	13.534545	192.168.2.1	192.168.2.18	DNS	Standard query response A 128.100.11.1
4	13.541026	192.168.2.18	128.100.11.1	ICMP	Echo (ping) request
5	13.555913	128.100.11.1	192.168.2.18	ICMP	Echo (ping) reply
6	14.542842	192.168.2.18	128.100.11.1	ICMP	Echo (ping) request
7	14.567211	128.100.11.1	192.168.2.18	ICMP	Echo (ping) reply
8	15.547669	192.168.2.18	128.100.11.1	ICMP	Echo (ping) request
9	15.586209	128.100.11.1	192.168.2.18	ICMP	Echo (ping) reply
10	16.552528	192.168.2.18	128.100.11.1	ICMP	Echo (ping) request
11	16.565526	128.100.11.1	192.168.2.18	ICMP	Echo (ping) reply
12	22.941534	192.168.2.18	192.168.2.255	BROWSER	Domain/workgroup Announcement @HOME, windows for wor

Frame 4 (74 bytes on wire, 74 bytes captured)

- Ethernet II, Src: 00:01:03:1d:cc:f7, Dst: 00:04:e2:29:b2:3a
- Internet Protocol, Src Addr: 192.168.2.18 (192.168.2.18), Dst Addr: 128.100.11.1 (128.100.11.1)
- Internet Control Message Protocol
 - Type: 8 (Echo (ping) request)
 - Code: 0
 - Checksum: 0xf05b (correct)
 - Identifier: 0x0200
 - Sequence number: 5b:00
 - Data (32 bytes)

Raw packet data (hex and ASCII):

```
0000  00 04 e2 29 b2 3a 00 01 03 1d cc f7 08 00 45 00  ...).:... ..E.
0010  00 3c 19 8a 00 00 20 01 33 18 c0 a8 02 12 80 64  .<.... 3.....d
0020  0b 01 08 00 f0 5b 02 00 5b 00 61 62 63 64 65 66  ....[.. [..abcdef
0030  67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76  ghijklmn opqrstuv
0040  77 61 62 63 64 65 66 67 68 69                    wabcdefg hi
```

Example – Echo Reply

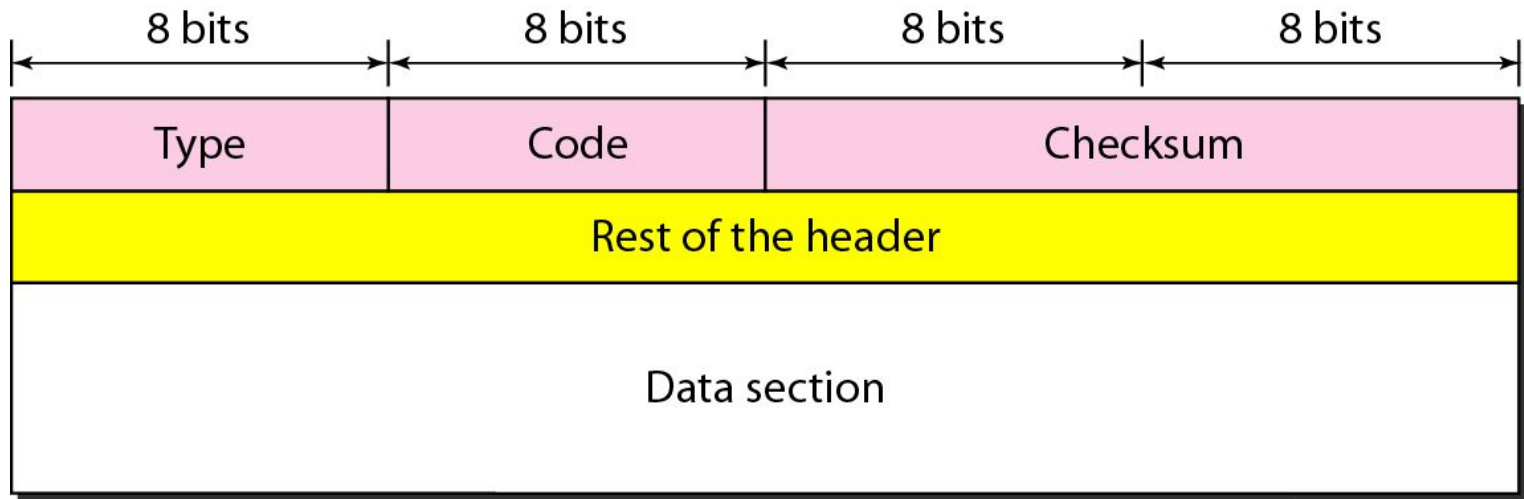
The image shows a Wireshark packet capture window titled "pingtesla - Ethereal". The main pane displays a list of 11 network packets. Packet 5 is selected, showing it is an ICMP Echo (ping) reply from 128.100.11.1 to 192.168.2.18. The details pane below shows the structure of this packet: Ethernet II, Internet Protocol, and Internet Control Message Protocol (Type: 0, Code: 0, Checksum: 0xf85b, Identifier: 0x0200, Sequence number: 5b:00). The packet data is shown in hexadecimal and ASCII at the bottom.

No.	Time	Source	Destination	Protocol	Info
1	0.000000	00000000.0001031dccf7	00000000.Broadcast	IPX SAP	Nearest Query
2	13.526454	192.168.2.18	192.168.2.1	DNS	Standard query A tesla.comm.utoronto.ca
3	13.534545	192.168.2.1	192.168.2.18	DNS	Standard query response A 128.100.11.1
4	13.541026	192.168.2.18	128.100.11.1	ICMP	Echo (ping) request
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6	14.542842	192.168.2.18	128.100.11.1	ICMP	Echo (ping) request
7	14.567211	128.100.11.1	192.168.2.18	ICMP	Echo (ping) reply
8	15.547669	192.168.2.18	128.100.11.1	ICMP	Echo (ping) request
9	15.586209	128.100.11.1	192.168.2.18	ICMP	Echo (ping) reply
10	16.552528	192.168.2.18	128.100.11.1	ICMP	Echo (ping) request
11	16.565526	128.100.11.1	192.168.2.18	ICMP	Echo (ping) reply

Frame 5 (74 bytes on wire, 74 bytes captured)
Ethernet II, Src: 00:04:e2:29:b2:3a, Dst: 00:01:03:1d:cc:f7
Internet Protocol, Src Addr: 128.100.11.1 (128.100.11.1), Dst Addr: 192.168.2.18 (192.168.2.18)
Internet Control Message Protocol
Type: 0 (Echo (ping) reply)
Code: 0
Checksum: 0xf85b (correct)
Identifier: 0x0200
Sequence number: 5b:00
Data (32 bytes)

```
0000  00 01 03 1d cc f7 00 04 e2 29 b2 3a 08 00 45 00  .....).:..E.  
0010  00 3c 99 88 00 00 f0 01 e3 18 80 64 0b 01 c0 a8  .<.....d....  
0020  02 12 00 00 f8 5b 02 00 5b 00 61 62 63 64 65 66  .....[. [.abcdef  
0030  67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76  ghijklmn opqrstuv  
0040  77 61 62 63 64 65 66 67 68 69                    wabcdefg hi
```

General format of ICMP messages



Note

ICMP always reports error messages to the original source.